Complex Network Analysis

Analysis of the Chemical–Gene Interaction Dataset
(ChG-InterDecagon, ID 10016)

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Contents

1 Introduction

The analysis of interactions between drugs and genes represents a very important topic in computational pharmacology. These analyses make it possible to explore new interaction mechanisms or to re-evaluate those already widely used. The ChG-InterDecagon dataset makes it possible to investigate not only drug–gene interactions already known in the literature, but also new interactions, since it aggregates information from different sources such as experiments carried out under different conditions and associations derived from computational models. In this work, the dataset is analyzed through an approach based on network theory with the objective of characterizing the global topological structure and the local functional organization of the system.

2 Objective

The general objective of the project is to analyze the topological and functional structure of the Chemical–Gene Interaction Network (ChG-InterDecagon) dataset by examining how drugs, grouped and distributed according to defined similarity criteria, are organized and act on the different genes present in the dataset. After constructing and visualizing a representation of the bipartite drug–gene network, the aim is to build a similarity network that identifies drugs with similar profiles through Jaccard similarity. Through this network, parameters such as density and modularity, which describe the internal organization of the drug network, are analyzed.

The second objective is to analyze how, through the Louvain method, drugs organize into communities in order to identify whether they are homogeneous or heterogeneous (dividing into subgroups) and whether different communities interact with each other.

Subsequently, the focus shifts to the genes, aiming to analyze co-targeting and its redundancy within the communities identified previously. In this way, it is possible to identify strongly co-targeted genes and internal functional modules. This may have interesting biological implications.

Since parameters such as density and clustering coefficient do not allow for a deeper characterization of the internal structure of the communities, a spectral analysis using the normalized Laplacian is carried out. This makes it possible to evaluate global structural cohesion and the possible presence of modular substructures.

As a final objective, relationships of generality/specificity among drugs within the same communities are analyzed through the construction of a Directed Acyclic Graph (DAG). This makes it possible to identify maximal gene profiles, incremental variants, and specific profiles, thereby providing an initial insight into the pharmacological structure of the community.

3 Dataset description

The report analyzes the dataset "Chemical-gene interaction network" (ID 10016-ChG-InterDecagon), which can be found at the following link: `https://snap.stanford.edu/biodata/datasets/10016/10016-ChG-InterDecagon.html`. It contains a biological network in which the nodes are drugs/chemical compounds and genes/proteins. The edges of the network represent the biological interactions between these elements. Such interactions are functional or biomedical associations, such as the binding of a chemical compound to a target protein, the activation or inhibition of a gene, experimentally observed effects, and computational predictions. For the subsequent analyses, it is important to highlight that the dataset is not limited to a single context, but also aggregates data coming from different experimental conditions. Below is a table with all the specific information regarding the dataset:

Dataset statistic	**Valore**
Nodes	9 569
Drug nodes	1 774
Gene nodes	7 795
Edges	131 034
Nodes in largest SCC	9 538
Fraction of nodes in largest SCC	1.000
Edges in largest SCC	131 001
Fraction of edges in largest SCC	1.000
Diameter (longest shortest path)	8
90-percentile effective diameter	3.864

Table 1: Dataset statistics of the ChG-InterDecagon network.

Source: https://snap.stanford.edu/biodata/datasets/10016/10016-ChG-InterDecagon.html

4 Pre-processing

Before any analysis and the creation of the networks, a preprocessing of the data in the dataset was carried out. After an initial syntactic cleaning (removal of comments, empty spaces, and incomplete rows), normalization was performed in order to obtain well-identified genes IDs and drugs that could be easily used for the subsequent analyses. In particular, the identifiers were cleaned by removing empty spaces and prefixes, followed by the conversion of the identifier from a string to a numerical value. Furthermore, all duplicate rows were removed in order to avoid having "artificial" nodes in the dataset.

After preprocessing, the drug identifiers and the gene IDs have the following form:

Drug ID	**Gene ID**
60 752	3 757.000
6 918 155	2 908.000
103 052 762	3 359.000
23 668 479	1 230.000
28 864	1 269.000

Table 2: Example of drug and gene IDs after preprocessing.

5 Methodology

5.1 Bipartite graph

For a visual representation of the network, a bipartite drug–gene graph was created. In order to clearly visualize the nodes, not all of them were used, but only those with a degree between 5 and 15, imposing a maximum number of 50 drug nodes out of a total of 200 nodes. The resulting graph has:

Nodes	Edges
137	403

Table 3: Number of nodes and edges in the graph.

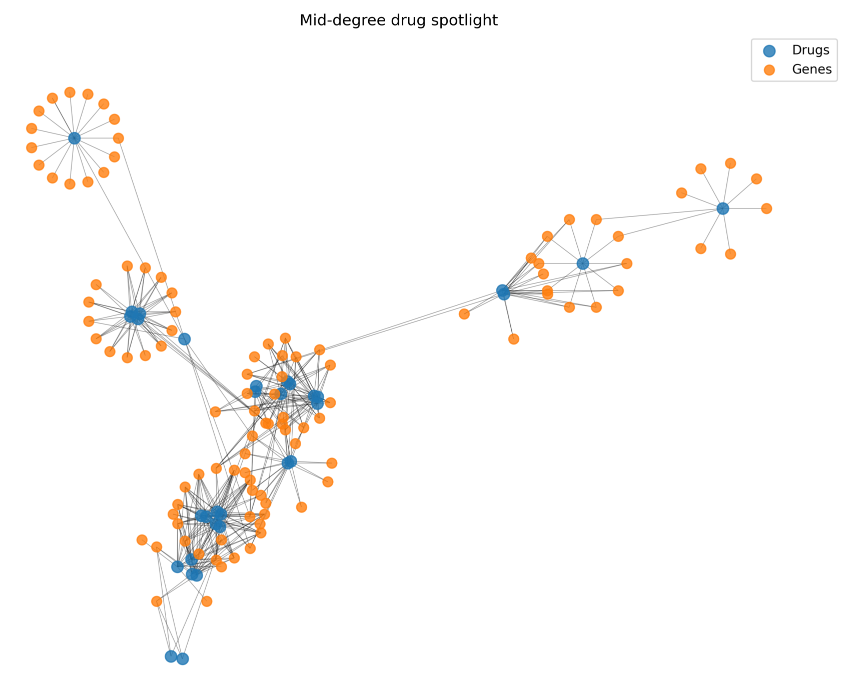

Figure 1: Example drug gene bipartite graph

5.2 Similarity network

A drug–drug similarity network was constructed using Jaccard similarity, initially applying a threshold equal to 0.3 and subsequently one equal to 0.4. The results obtained with the two thresholds were then compared with each other.

The Jaccard similarity measures the degree of relative overlap between two drugs and is defined as:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

It indicates that two drugs are considered similar only if they share a significant number of targets and, at the same time, have few different targets, that is, if they have an overall similar target profile.

Jaccard similarity was chosen because it operates natively on sets, penalizing drugs with a high number of non-shared targets. This makes it particularly suitable for analyzing similarity between target profiles typical of the dataset under study.

Each node of the similarity network graph represents a drug, characterized by a target profile, that is, by a vector containing all the genes (targets) with which the drug interacts. An example of the representation of a node is the following:

Sposta all'inizio del paragrafo similarity

```
{
  "bipartite": "drug",
  "original_id": 12345,
  "targets": [1017, 1956, 7422]
}
```

5.2.1 Threshold 0.3

The parameters related to the filtering of the similarity network with threshold 0.3 are the following:

Parameter	Value
Similarity threshold	0.300
Nodes removed	333
Edges filtered	961 999
Original node count	1 774
Retained node count	1 441
Potential edges	1 037 520

Table 4: Similarity network parameters (threshold = 0.3).

This threshold value indicates that an edge (drug–drug) is present in the network only if the similarity between two drugs is ≥ 0.3. It is a moderate threshold that retains connections with medium–low similarity, thus preserving a relatively dense network compared to more stringent thresholds. It generates a connected but not excessively connected network, useful for community detection, reducing noise and eliminating extremely weak similarities.

Since the number of original nodes (*original_node_count*) is 1774 and the number of "retained" nodes (*retained_node_count*) is 1441, it is observed that, after filtering, the network retains 81.2% of the drugs.

The parameter *potential_edges* represents the total number of possible drug–drug pairs before filtering, while *edges_filtered* indicates how many pairs were removed because they had similarity < 0.3. By subtracting these two values, it is obtained that the final network has 85,310 edges. Therefore, only about 8.22% of the possible connections exceed the threshold, indicating a relatively sparse network. This sparsity is consistent with pharmacological similarity networks; in fact, generally only a few drugs are truly similar.

5.2.2 Threshold 0.4

The parameters related to the filtering of the similarity network with the threshold set to 0.4 are the following:

Parameter	Value
Similarity threshold	0.400
Nodes removed	333
Edges filtered	963 962
Original node count	1 774
Retained node count	1 441
Potential edges	1 037 520

Table 5: Similarity network parameters (threshold = 0.4).

With a threshold of 0.4, the threshold is more stringent, retaining only stronger interactions than before. A total of 963,962 pairs are filtered out, while 73,558 out of 1,037,520 possible pairs are retained. The percentage of connections that manage to exceed the threshold is 7.56

With this threshold, the network loses $85.310 - 78.403 = 6.907$ edges compared to the previous case. This number represents a decrease of

$$\frac{6.907}{85.310} \approx 8.1\%$$

in the number of connections compared to the previous threshold.

The number of removed nodes remains unchanged at 333. This indicates that these drugs do not exhibit similarity values equal to or greater than 0.4 with any other drug in the dataset, thus already appearing as isolated in the similarity network at the previous threshold.

One of the aims of this analysis is to construct a community network in order to identify communities of drugs with similar mechanisms of action. In this context, raising the threshold too much would not make sense, as it could have the effect of retaining only the "obvious" relationships and losing weaker but biologically interesting ones. If two drugs share even only partially a set of genes, they may have similar side effects and potentially interesting implications for future analyses. A threshold that is too high risks eliminating these cases, retaining only obvious and already well-studied relationships.

The following analyses were performed considering a threshold of 0.4.

5.2.3 Similarity network global data

Density measures the fraction of pairs of drugs whose Jaccard similarity exceeds the chosen threshold. It therefore represents an indicator of pharmacological redundancy at the drug profile level.

The produced network has 1,441 total nodes, 73,558 edges, and a density equal to 0.07; that is, 93% of drug pairs do not exceed the Jaccard threshold. This indicates a globally sparse network, yet a value typical of biological similarity networks. Density was calculated as:

$$\text{density} = \frac{2E}{N(N-1)}$$

As can be observed from the following representation of the similarity network, and as will be discussed later, the global density is strongly influenced by the presence of giant components, in particular by the one with a size of 359 nodes.

Below is a representation of 500 nodes randomly selected from the similarity network.

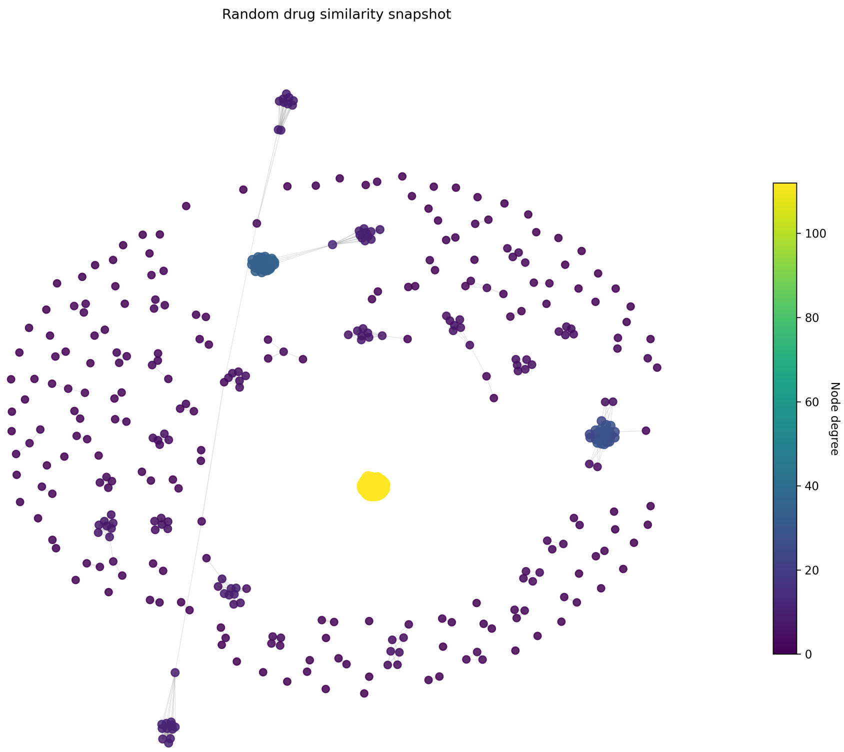

Figure 2: Example drug gene bipartite graph

5.3 Community network analysis

After constructing the similarity network, the corresponding community network was built using the Louvain method. Below are the Louvain parameters saved in the file *louvian_parameters.json* after the execution of the analysis.

Parameter	Value
Method	Louvain
Resolution	1.000
Modularity	0.204
Minimum community size	1
Maximum community size	359
Mean community size	5.521
Median community size	2.000

Table 6: Community detection parameters and global statistics (Louvain method).

5.3.1 Parameters

- Modularity: A modularity of approximately 0.20 indicates the presence of a weak but real modular structure. This suggests that the drugs do not have a random distribution but "organize" into communities, that is, groups characterized by partially shared gene profiles. As will be discussed later, genes belonging to different communities can be connected to each other, producing (at least in one case) an interconnected structure.
- Median: The median value of the communities is 2. This means that more than half of the communities contain 2 drugs or fewer. This suggests the presence of pharmacological groups that are very similar to each other, which may be structural variants of the same compound or drugs that share one or very few highly specific targets. In other words, these communities represent niche drugs with rare targets or may be interesting outliers for repositioning.
- Mean: The mean corresponds to a community of approximately 5 drugs. This value, together with the median, indicates a strongly skewed distribution with a few very large clusters and many small ones.
- Max: The maximum value corresponds to a community with 359 elements. This community will be analyzed in greater detail in the following sections.

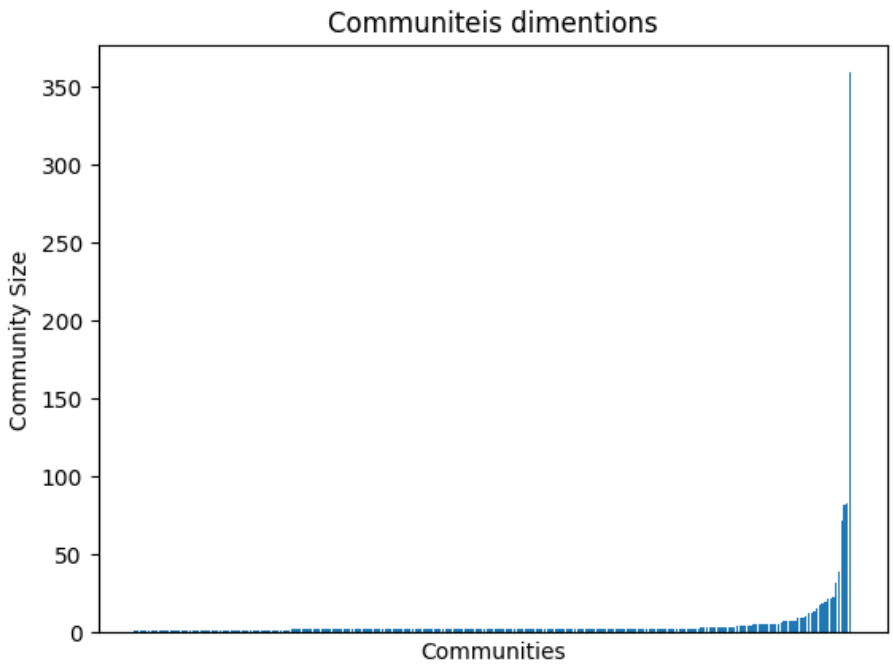

Figure 3: Representation of the sizes the sizes of all the communities

5.4 Communities analysis

The individual communities are saved in the file *community_parameters.csv*. The file is organized into columns, each corresponding to a characteristic of the community:

- community id: Identifier of the community
- size: Size of the community
- degree: Number of edges of the node
- weighted degree: Sum of the weights of the edges incident to a node
- clustering coefficient: Degree of local closure of the communities

In the following table, the data obtained from the analysis of only the communities with size greater than or equal to 10 are reported.

Community	**Size**	**Degree**	**Weighted Degree**	**Clustering**	**Density**
Community_5	359	0	0.000	0.903	1.000
Community_13	18	0	0.000	0.000	0.791
Community_20	22	0	0.000	0.593	0.792
Community_21	82	0	0.000	0.631	0.778
Community_22	83	1	12.990	0.851	0.913
Community_53	10	0	0.000	0.000	0.733
Community_54	22	0	0.000	0.574	0.745
Community_58	11	0	0.000	0.000	1.000
Community_59	10	0	0.000	0.000	0.889
Community_78	32	1	12.990	0.653	0.905
Community_81	20	0	0.000	0.964	1.000
Community_97	19	0	0.000	0.000	0.754
Community_100	13	0	0.000	0.000	0.346
Community_109	13	0	0.000	0.000	1.000
Community_127	14	0	0.000	0.000	0.604
Community_135	16	0	0.000	0.000	0.425
Community_187	10	0	0.000	0.000	1.000
Community_188	72	0	0.000	0.616	0.324
Community_192	39	0	0.000	0.706	0.601
Community_203	23	0	0.000	0.513	0.482

Table 7: Structural parameters of selected communities in the similarity network.

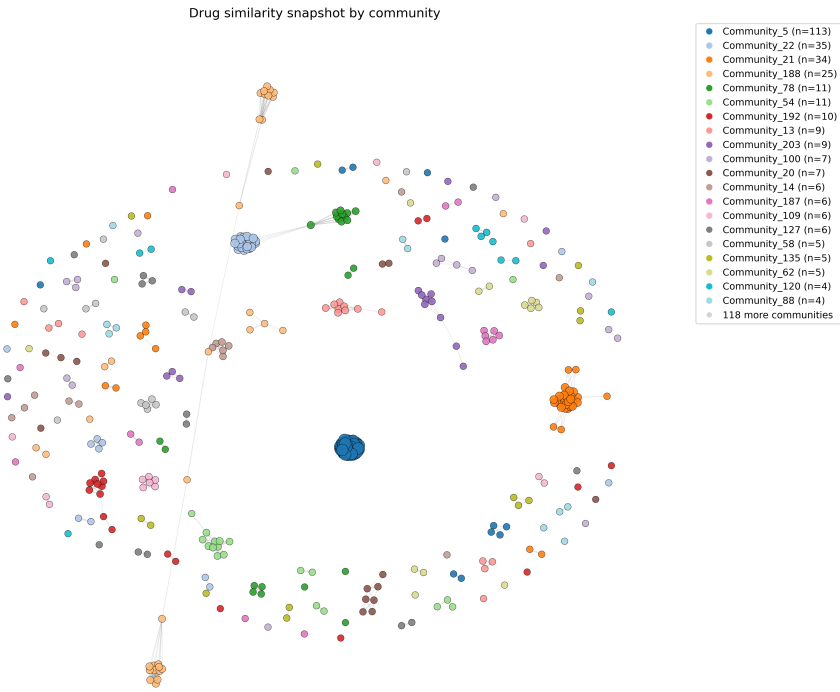

Figure 4: Graphic representation of the communities

5.4.1 Size

The distribution of community sizes indicates the presence of a high number of small pharmacological modules. These drugs may have rare or highly specific profiles or may represent mechanisms of action that are poorly redundant or scarcely explored and, with further analysis, may serve as interesting starting points for the study of new drugs.

The presence of very large communities can also be observed. These may correspond to areas associated with widely studied targets or pathways. Another plausible reason for their presence may be, as will be discussed later, the intrinsic nature of the dataset. Indeed, it contains data obtained through variants of the same experimental condition or under different conditions of the same compound.

Since these drugs act on very similar gene sets, more in-depth analyses could be conducted to investigate possible therapeutic combinations, including among drugs belonging to closely connected communities.

Size range	**Count**	**Percent (%)**
Size < 5	219	83.910
$5 \leq$ Size ≤ 50	38	14.560
$50 <$ Size ≤ 100	3	1.150
Size > 100	1	0.380

Table 8: Distribution of community sizes in the similarity network.

5.4.2 Density

Density measures how similar the drugs within a community actually are. In this context, density helps distinguish which communities are truly homogeneous from those whose components share only some targets.

$$\text{density} = \frac{E_{\text{int}}}{\frac{n(n-1)}{2}}$$

where:

- density: fraction of internal connections present with respect to the maximum possible within the community.
- *E_ int*: number of internal edges within the community (only between nodes of the same community).
- n: number of nodes in the community (size).
- $\frac{n(n-1}{2}$: maximum number of possible edges in an undirected community without self-loops.

In small modules, the number of possible connections is very limited; in fact, it is sufficient for a few nodes to be all connected to each other for the density to be high, often close to 1. This reflects a very strong similarity among the drugs in the group (for example, almost complete sharing of targets), but such values are not very robust from a statistical point of view, because they are strongly influenced by the low number of nodes. For this reason, they were removed from the analysis, considering only communities with size greater than or equal to 10.

From the reported data, it can be observed that medium–large communities exhibit variable density values. Analyzing the communities with size between 39 and 83, very different densities can be observed, namely values ranging between 0.324 and 0.913. The causes of this behavior may be both mathematical and biological.

The mathematical causes are attributable to how density is calculated (previous formula). Since the number of possible internal connections increases quadratically with the size of the community, even small variations in node profiles tend to accumulate into an increasing number of missing edges. This effect is amplified in larger communities, where the high number of profile comparisons makes the emergence of differences more likely, resulting in greater variability in density values.

Moreover, the presence or absence of an edge depends on a threshold (0.4). Therefore, if two nodes are just below the threshold, the edge in the similarity network is absent, whereas if the similarity is just above the threshold, the edge is present. This introduces artificial variability and greater dispersion in the density profiles.

From a biological perspective, large communities often aggregate broad pharmacological families, complex pathways, or partially overlapping targets, consequently creating larger clusters that are not completely connected, which is reflected in lower and more variable density values.

For smaller communities (size between 10 and 32), a generally high density is observed, with most values ranging between 0.7 and 1.0. However, some exceptions with more moderate or low density are present, such as Community 135 (0.425), Community 203 (0.482), and Community 100 (0.346).

5.4.3 Clustering coefficient

For each community, the clustering coefficient was calculated:

Community	Size	Density	Clustering Coefficient
Community_5	359	1.000	0.903
Community_13	18	0.791	0.746
Community_20	22	0.792	0.593
Community_21	82	0.778	0.631
Community_22	83	0.913	0.851
Community_53	10	0.733	0.660
Community_54	22	0.745	0.574
Community_58	11	1.000	0.831
Community_59	10	0.889	0.629
Community_78	32	0.905	0.653
Community_81	20	1.000	0.964
Community_97	19	0.754	0.553
Community_100	13	0.346	0.442
Community_109	13	1.000	0.787
Community_127	14	0.604	0.671
Community_135	16	0.425	0.596
Community_187	10	1.000	0.660
Community_188	72	0.324	0.616
Community_192	39	0.601	0.706
Community_203	23	0.482	0.513

Table 9: Density and clustering coefficient of selected communities in the similarity network.

Except for a few cases, the clustering coefficient is always > 0.5. In these cases, it can be said that such communities do not exhibit a linear or "chain-like" structure, in which drugs are similar only to a few immediate neighbors, but rather form strongly cohesive modules. That is, if a drug is similar to other drugs within the community, it is highly likely that, given how similarity was defined and the threshold used, it is also similar to other components of the same community.

Communities 13, 20, 21, 22, 53, 54, 59, 78, 97 exhibit values within the following range:

Size	Density	Clustering Coefficient
10–83	0.733–0.913	0.553–0.851

Table 10: Range of size, density, and clustering coefficient for medium-sized communities.

therefore with high density but lower than 1 and moderately high or high clustering. From these data, it can be stated that not all drugs are similar to each other, but that highly coherent subgroups exist within the community. Biologically, this may reflect the presence of pharmacological families in which there is a core of common targets and in which each drug may introduce marginal variations in the profile. There are also communities with low density and moderate clustering.

Among the communities with lower density, we have:

Community	Size	Density	Clustering Coefficient
Community_100	13	0.346	0.442
Community_135	16	0.425	0.596
Community_188	72	0.324	0.616
Community_203	23	0.482	0.513

Table 11: Communities with relatively low density values in the similarity network.

These show few global connections but well-structured local connections, highlighting the fact that the community is not a compact block but rather a set of local clusters indirectly connected.

A final case that can be observed from the previous table consists of communities with density ≈ 1 and high clustering.

Community	Size	Density	Clustering Coefficient
Community_5	359	1.000	0.903
Community_58	11	1.000	0.831
Community_81	20	1.000	0.964
Community_109	13	1.000	0.787
Community_187	10	1.000	0.660

Table 12: Communities with unit density (clique-like structures) in the similarity network.

In these communities, the clustering coefficient does not add new information with respect to density, but reinforces the evidence of extreme homogeneity.

5.4.4 Weighted degree

To describe the overall level of interaction of a community with the others, the weighted degree was calculated. It is computed as the sum of the weights of the inter-community edges (Jaccard similarity). Only communities 23 and 78 are connected to each other with a weighted degree = 12.99.

Since an edge between two communities exists only if there are drugs belonging to different communities that are connected and therefore share a significant portion of gene targets (Jaccard greater than or equal to 0.4), the two communities represent distinct but not independent pharmacological modules. The "bridge" drugs that connect the two communities may act on different pathways that are nevertheless interconnected through key genes. Such drugs may have potential common side effects and possible opportunities for therapeutic combination or drug repurposing.

5.4.5 Clique community

Two almost-clique communities are present in the network. The first is the community with the largest number of elements. It exhibits the following characteristics:

- **density**: 0.999
- **size**: 359

- **unique_profiles**: 120
- **shared_profiles**: 81
- **most_frequent_profile**:
  - **drug_count**: 125
  - **gene_count**: 158

Therefore, more than one third of the components of the community share the exact same set of genes. This implies that the Jaccard similarity is $J = 1$ for all pairs of drugs sharing the same profile and $J \approx 1$ for pairs of nodes that differ from each other by only a few genes. Consequently, the subgraph of the community, that is, the graph containing the nodes of the community and all the corresponding similarity edges, is almost completely connected, producing an almost unitary density.

One possible cause of this high density value for such a large community may be that Jaccard becomes very permissive when the sets are large and the intersection between two sets is very wide. In fact, when dealing with large sets, even with dozens of different genes, the similarity remains high. Moreover, with a sufficiently permissive threshold, as in our case, the value of $E_{\text{int}} \approx \frac{n(n-1)}{2}$, resulting in density ≈ 1.

A further cause of the almost unitary density value of this community may arise from the intrinsic nature of the dataset. It does not only aggregate information coming from different sources (real screenings, experimental data, computationally generated results), but also information about the same compound obtained from different screenings. This, combined with the fact that the drug IDs within the dataset do not distinguish the experimental context in which the data were produced, may lead to the generation of drugs with identical or nearly identical profiles.

Without any further analysis, this community represents a family of drugs with an almost identical mechanism of action.

There are other communities with unitary density to which the same reasoning just discussed can be applied. In addition to this, it should be considered that trivial cliques are also present, as in the case of Community 20. It contains 20 drugs, 19 of which have the exact same profile composed of only two genes. This represents a more trivial clique than the previous one, since the profile is extremely small and perfectly replicated.

5.5 Co-occurence network

To analyze the frequency with which pairs of genes appear together in drug profiles, it was decided to construct a series of co-occurrence networks. Due to the high computational cost associated with calculating a global co-occurrence network, it was chosen to create and analyze co-occurrence networks for each community previously identified through community analysis. The focus was placed on communities with size ≥ 15. In addition to this filter, another was applied in order to make the data easier to interpret by removing noise and quasi-clique structures. In particular, super-frequent genes were removed from the communities, namely genes whose number of associated drugs is greater than the 95th percentile of the "drugs per gene" distribution.

Below are the parameters of the co-occurrence networks obtained for each community:

Community	Nodes	Edges	Density	Components	Giant Comp.	Clustering	Comm. Size
Community_5	357	47 957	0.755	1	357	0.819	359
Community_13	166	13 626	0.995	1	166	0.997	18
Community_20	17	136	1.000	1	17	1.000	22
Community_21	26	280	0.862	1	26	0.910	82
Community_22	77	2 563	0.876	1	77	0.914	83
Community_54	4	6	1.000	1	4	1.000	22
Community_78	89	3 394	0.867	1	89	0.921	32
Community_81	2	1	1.000	1	2	0.000	20
Community_97	215	22 151	0.963	1	215	0.973	19
Community_135	86	2 863	0.783	1	86	0.883	16
Community_188	82	2 594	0.781	1	82	0.865	72
Community_192	34	427	0.761	1	34	0.872	39
Community_203	5	10	1.000	1	5	1.000	23

Table 13: Co-occurrence network structural parameters for selected communities.

5.5.1 Component count and giant component size

All gene–gene co-occurrence networks have component count = 1. This holds regardless of the number of nodes (from 2 to 357), density (from ≈ 0.75 to 1), and clustering coefficient. This indicates that there are no isolated genetic subgroups within the community, that is, the gene–gene network is fully connected. In this context, every gene is reachable from every other gene through at least one path.

Taking into account the limits imposed by the analysis and by the intrinsic nature of the dataset (discussed earlier), from a biological perspective, a component count value of 1 suggests that the genes within each community participate in an interconnected functional system. No separate genetic subsets emerge, but rather a single functional block.

Since the global connectivity of the network is already guaranteed (component count = 1) by construction, in this case density measures the fraction of gene pairs that share at least one drug within the community. It therefore represents an indicator of pharmacological redundancy.

Density, calculated using the formula presented earlier, shows high values in almost all cases, that is, there are no sparse networks (density > 0.5). Two different regimes can be identified, although all with high values.

In the first density regime, values lie in the range [0.95, 1.00]. In this case, there is an almost total absence of profile heterogeneity, where all or almost all pairs of genes are connected (clique or quasi-clique networks). There is strong pharmacological redundancy, in which genes are targeted, in most cases, by the same drugs (strong pharmacological redundancy).

In the second regime, density ranges between [0.75, 0.90). We are in the presence of networks that are still fully connected but with a more or less significant number of missing edges. In these structures, which are less clique-like and more articulated than the previous ones, there may be genes involved in distinct processes but indirectly interconnected.

Also in the case of the clustering coefficient, very high values are observed. Two regimes can be distinguished: maximum values and high values but < 1. In the first case, we observe genetically totally redundant structures in which every triple forms a triangle. In the second case ($\approx 0.88\breve{}0.95$), we have structures that are not completely closed, with more articulated genetic modules compared to the previous case, with the presence of bridge genes connecting different functional subsets.

There is one community with clustering coefficient = 0. This is a network composed of 2 genes, therefore with no triangles by definition.

5.5.2 Wheight distribution and sparsity

Community	**Nodes**	**Median**	**Mean**	**Max**	**Weight = 1 (%)**	**Weight = 2 (%)**
Community_5	880	2.000	32.209	359	6.432	64.565
Community_13	453	2.000	3.016	18	14.154	65.961
Community_20	38	2.000	3.701	20	10.598	71.467
Community_21	52	2.000	3.812	78	31.768	35.368
Community_22	181	2.000	8.426	83	9.834	62.845
Community_54	14	2.000	2.588	21	0.000	94.118
Community_78	220	2.000	4.178	31	29.849	39.737
Community_81	3	1.000	7.333	20	66.667	0.000
Community_97	390	2.000	4.371	19	11.056	49.854
Community_135	175	2.000	2.573	9	46.854	30.391
Community_188	130	2.000	5.874	33	6.186	45.450
Community_192	63	2.000	5.691	31	1.700	71.000
Community_203	14	2.000	2.730	17	21.622	56.757

Table 14: Weight distribution and sparsity metrics of the co-occurrence networks for selected communities.

Observing the data in the table, it can be seen that the median is equal to 2 in all communities except for Community 81, which represents a boundary case having 3 nodes. Moreover, in general, in all communities, mean > median is observed, indicating asymmetric distributions (with a long right tail), characterized by the presence of a few edges with high weight and many edges with low weight. In some communities (for example Community 5) this difference between mean and median is more pronounced, while in others it is more limited (for example Community 20). This suggests the presence of pairs of genes that are strongly co-targeted, embedded in a context of overall more moderate co-targeting.

The parameter max measures the maximum weight observed in the community, that is, the maximum number of drugs shared by a pair of genes. In other words, it is an indicator of how strong co-targeting can be in that community. There are many communities with a high max value, indicating that in all networks there exists at least one pair of genes co-targeted by a large number of drugs.

The parameters *weight_ eq_ 1_ pct* and *weight_ eq_ 2_ pct* were introduced as additional information and represent, respectively, the proportion of gene–gene edges supported by one and by two drugs. It can be observed that *weight_ eq_ 1_ pct* is always well below 50% (with the exception of very small-sized cases), while *weight_ eq_ 2_ pct* is often the dominant class (≈ 40˘70%). This behavior is consistent with a median weight equal to two, which implies that at least half of the edges present in the co-occurrence networks are supported by two or more drugs.

5.6 Spectral analisys

In order to encode the internal connectivity structure of each drug community, the normalized Laplacian matrices were constructed and, from them, the Fiedler value was analyzed.

Although, in a previous analysis, parameters such as density and clustering coefficient were calculated, providing an initial topological characterization of the communities, the spectral analysis of the normalized Laplacian makes it possible to evaluate global structural cohesion and the possible presence of modular substructures not detectable with the aforementioned metrics.

Density, for example, is a global characterization indicating how many edges are present in the community. Indeed, two communities with the same density value may be structurally different (a quasi-clique community or one consisting of two dense separated blocks). In this context, it is difficult to distinguish truly homogeneous communities from aggregations induced by the similarity measure. To analyze network connectivity, the Fiedler values, that is, the second smallest eigenvalues, were calculated.

To perform an analysis of the identified Fiedler values, we introduce the following operational classes:

λ_1 **Range**	**Structural Interpretation**
$\lambda_1 \ll 0.1$	Community **nearly separable**, fragile structure
$0.1 \leq \lambda_1 \leq 0.4$	Marked internal modularity
$0.4 \leq \lambda_1 \leq 0.8$	Structurally cohesive with internal heterogeneity
$0.8 \leq \lambda_1 \leq 1.1$	Community **strongly cohesive**
$\lambda_1 > 1.1$	Near-clique / extremely compact structure

Table 15: Structural interpretation of the Fiedler value (λ_1) in community spectral analysis.

In the dataset, there are both spectrally weak communities (e.g, Community 100, 135, 188) and spectrally very strong communities with $\lambda\_1 \approx 1$ or greater.

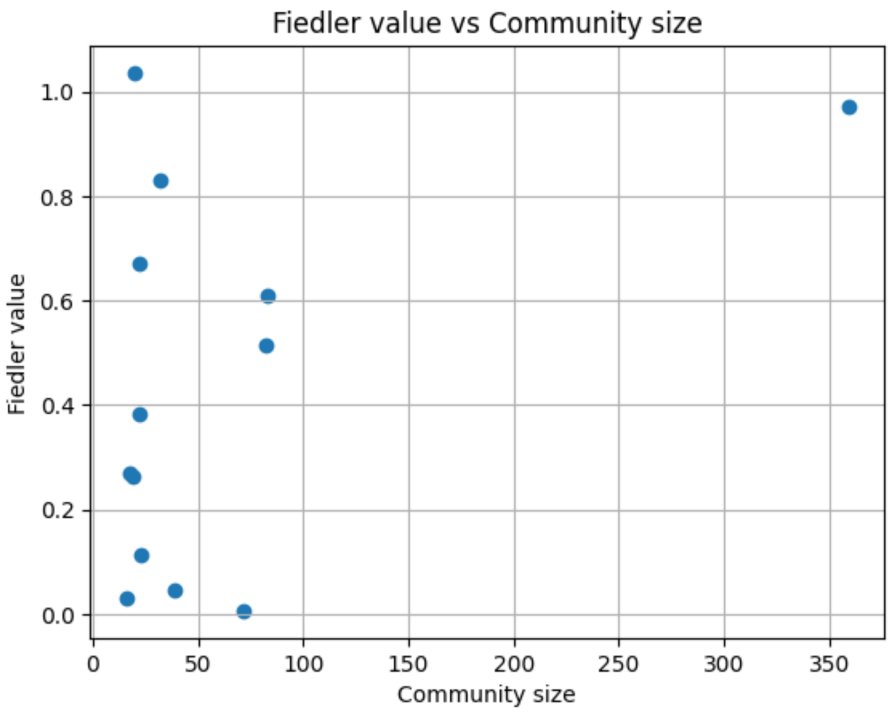

Figure 5: Fiedler value in function of the size

From the graph, it can be observed that, for smaller communities, the Fiedler value can take both small and large values, identifying communities with high connectivity as well as others with lower connectivity. As for larger communities (size > 50), with the exception of one case, they appear to be more connected.

In summary, density provides information only about how many edges are present, but not about how they are distributed or whether internal separations exist, whereas the Fiedler value provides information about how easy or costly it is to separate the graph into two parts, minimizing the weight of the cut edges (tutorial "Algorithms for Graph Partitioning"). Based on this, if density is high (many edges in the network) and the Fiedler value is high (difficulty in separating the graph), this indicates the presence of real redundancy, since edges are uniformly distributed, no separable subgroups are present, and each node is "well" connected to all others. Conversely, if the Fiedler value is low, it is "easy" to separate the graph; we are in the presence of artificial aggregation, that is, a community that appears dense and compact due to mathematical construction, but does not represent a biologically homogeneous set. We are in the presence of two (or more) internally very dense subgroups with few edges between them.

In the ChG-InterDecagon dataset, this occurs when Jaccard is permissive on large sets, and drugs share part of the targets but belong to different pathways. In this case, Louvain groups them together, so density remains high, but spectrally the community is not truly homogeneously connected. Below are the obtained data.

Community	**Size**	**Density**	λ_1 **(Fiedler Value)**
Community_13	18	0.791	0.270
Community_20	22	0.792	0.384
Community_21	82	0.778	0.514
Community_22	83	0.913	0.611
Community_5	359	1.000	0.971
Community_54	22	0.745	0.672
Community_78	32	0.905	0.830
Community_81	20	1.000	1.036
Community_97	19	0.754	0.263
Community_135	16	0.425	0.029
Community_188	72	0.324	0.006
Community_192	39	0.601	0.045
Community_203	23	0.482	0.112

Table 16: Comparison between density and Fiedler value (λ_1) for selected communities.

Resuming the analysis of the previous sections, since all the listed communities have high density ($\gtrsim 0.75$), the real discriminant is the Fiedler value. We can divide the communities (size ≥ 15) into 3 different groups:

1. Fiedler low ($\lambda_1 \lesssim 0.1$): In this range, communities with weak global cohesion, almost separable, are present. This group includes Communities 135, 188, and 192.

2. Fiedler intermediate ($0.1 < \lambda_1 < 0.7$): These communities are dense but structurally heterogeneous (communities with a certain internal modularity). This group includes Communities 13, 20, 21, 22, 97, and 203.

3. Fiedler high ($\lambda_1 \gtrsim 0.7$): Globally cohesive communities. This group includes Communities 5, 78, 54, and 2 (mathematically trivial). In this case, Community 5, previously analyzed, is not an artifact of Jaccard or of the threshold, but reflects real redundancy of the target profiles.

Below is an example of the eigenvalue spectrum for a non-clique community (selected according to the criteria explained later).

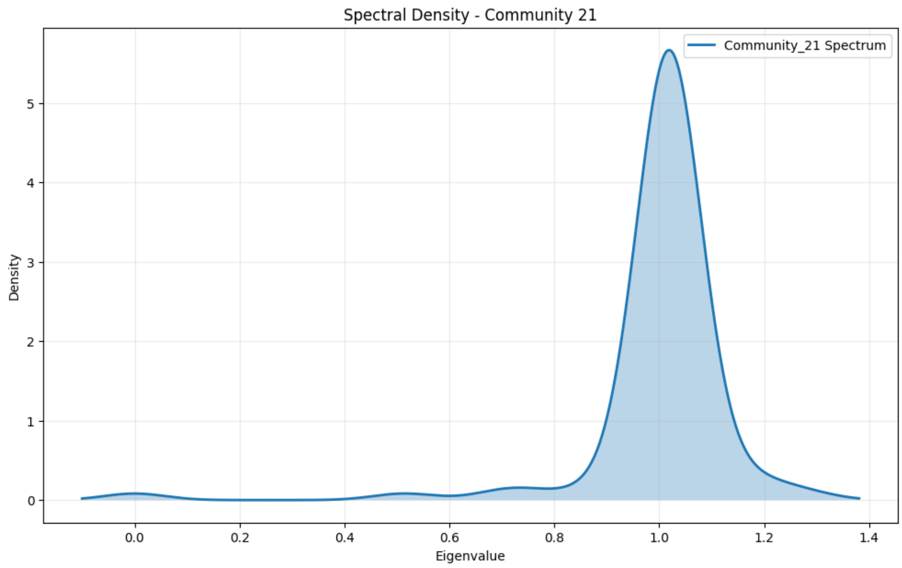

Figure 6: Eigen value spectra of community 21

5.7 DAG

One piece of information that communities do not provide concerns relationships of "generality"/"specificity" within communities of similar drugs. If two drugs are similar within a community, it is likely that they share a core set of genes and differ by a small periphery (a few additional or missing genes), meaning that there may exist more general drugs and more specialized drugs. The constructed DAGs, together with their orientation rules, and analyzed below, aim precisely to investigate this aspect.

5.7.1 Analyzed community

The following community was chosen for the analysis because it is an informative community, that is, neither trivial (small size) nor trivial in structure (clique or quasi-clique), and structurally heterogeneous and biologically interpretable node by node. The aim of the following analysis is to understand the role that each drug plays within the community. The community was chosen according to the following criteria.

- size $30 \leq \text{size} \leq 150$
- high density but not saturated: $0.7 \leq \text{density} \leq 0.9$
- high but not extreme clustering coefficient: $0.6 \leq \text{clustering} \leq 0.9$

- non-extreme Fiedler value: intermediate Fiedler (neither $\ll 0.1$ nor $\gg 0.8$)

The best candidate for this analysis is Community 21, which has the following parameters:

- size $= 82$
- density ≈ 0.78
- clustering ≈ 0.63
- Fiedler ≈ 0.51

5.7.2 Orientation rule

The DAG was constructed using the following orientation rule:

First, the targets are ordered by size. Subsequently, an edge is created from the larger set to the smaller set if the smaller set is a subset of the larger one ($A \to B$ if $T(B) \subset T(A)$). A constraint is imposed on the difference in cardinality, namely only pairs with $0 < |T(A)| - |T(B)| \leq$ `min_set_difference` (default 3) are considered, interrupting the loop if the difference is greater.

In this way, it follows that:

- A = more "general" profile (includes everything B does + extra)
- B = more "specific" profile (a subset of the general one)

The fact that A can have at most 3 additional targets compared to B is fundamental for a local hierarchical representation and in order not to lose biological interpretability.

The following types of nodes are identified:

- Sources: maximal drugs
- Sinks: minimal drugs
- Intermediate nodes: incremental variants

The global data of the DAG are the following:

Parameter	**Value**
Number of nodes	82
Number of edges	681
DAG density	0.103
Number of sources	35
Number of sinks	21
Maximum depth	6

Table 17: Global structural parameters of the DAG.

The observed density value suggests that the inclusion hierarchy among the nodes is present but not too dense, resulting in a partial representation of dominance relationships. This is consistent with the selective orientation rule used, which tends to connect only target profiles that are very similar in terms of size. In other words, density represents a biologically interpretable structure without inducing too many redundant or weakly informative connections.

The high number of sources (approximately 43% of the total nodes) is consistent with the constraint imposed on the difference between two "successive" sets. Consequently, many nodes with large target sets do not find subsets that are "close enough" and remain without incoming edges, becoming sources of the DAG. The number of sinks is lower.

Chains with a maximum depth of 6 are present, suggesting the possible presence of a process of progressive specialization starting from more "general" drugs and moving toward more "specific" ones.

By analyzing the high number of edges, it can be stated that many drugs differ by only a few genes and therefore often share a common core.

It is important to take into account the properties introduced by the orientation rule, since the obtained results are strongly influenced by it. Among these properties, we observe a local rather than global hierarchy imposed by the cardinality constraint, which also produces many sources.

5.7.3 In-degree and out-degree parameters

The following parameters were calculated for each node:

- *in_degree*: number of incoming edges
- *out_degree*: number of outgoing edges
- *degree_ratio*: $\dfrac{\text{out\_degree}}{\text{in\_degree} + 1}$
- **topological_level**: topological level (0 for sources, increasing along predecessors)

It was calculated that:

Degree Category	*in_degree* **Nodes**	*out_degree* **Nodes**
Equal to 0	35	21
Between 1 and 3	8	14
Greater than 5	39	47

Table 18: Distribution of nodes by in-degree and out-degree categories in the DAG.

35 drugs have *in_degree* $= 0$. These are not subsets of any other profile, being maximal profiles, and therefore may represent a common signature for many different drugs.

39 drugs have *in_degree* > 5, meaning they are nodes contained in many other profiles. These drugs may correspond to experimental variants or to different conditions of the same compound, as occurs for some drugs in the ChG-InterDecagon dataset.

Nodes with high outdegree represent hierarchical hubs, whereas those with zero outdegree are specific nodes that do not contain any others. This latter category includes both nodes that have never been connected to others (with no subset relationship) and the two sink nodes shown in yellow in the following graph.

5.7.4 degree ratio

The degree ratio was calculated as follows:

$$\text{degree\_ratio} = \frac{\text{out\_degree}}{\text{in\_degree} + 1}$$

It is a measure of the directional asymmetry of a node and is also an index of the hierarchical role of a node, since it can distinguish a node as:

- **Dominant role**: high degree ratio
- **Transition node**: $degree\_ratio \approx 1$
- **Leaf / specialized node**: low degree ratio

The calculated values are:

Statistic (*degree_ratio*)	**Value**
Mean	5.151 ± 1.181
Median	0.375 ± 0.375
Minimum	0.000
Maximum	31.000
Nodes with ratio < 1 (%)	73.171

Table 19: Descriptive statistics of the *degree_ratio* distribution in the DAG.

A strongly skewed distribution is observed, in which the majority of nodes have a value « 1 (leaf nodes), while only a few nodes exhibit very high values.

By observing the graph below, the presence of different roles can be identified: Isolated sources (purple at the top, $degree\_ratio = 0$) with profiles that cannot be ordered according to the chosen orientation rule. Dominant sources (purple at the top, $degree\_ratio$ » 1) with high out-degree. These are maximal target profiles and therefore roots of large sub-hierarchies. Intermediate routing nodes (dark blue / light blue in the center). The latter have:

- $in\_degree > 0$
- $out\_degree > 0$
- $degree\_ratio \approx 1$

They connect large profiles to more specific profiles. Shared subsets (turquoise, medium–low level) and pre-leaves (light green, penultimate level), which represent the bottlenecks of the hierarchy and are the last steps before maximal specialization, consequently having a very low degree ratio. The last identified type consists of pure leaves (yellow at the bottom, $degree\_ratio = 0$). These are the true terminals of the DAG, that is, the maximally specific profiles. They represent highly selective drugs or extremely specific experimental conditions.

Below is a summary of the hierarchy statistics and a visualization of the DAG:

Category	# Nodes	% of Total
Isolated sources	18	21.950
Dominant sources	17	20.730
Intermediate nodes	37	45.120
Shared subsets	6	7.320
Pre-leaves	2	2.440
Leaves	2	2.440
Total	**82**	**100.000**

Table 20: Hierarchical node classification identified in the DAG.

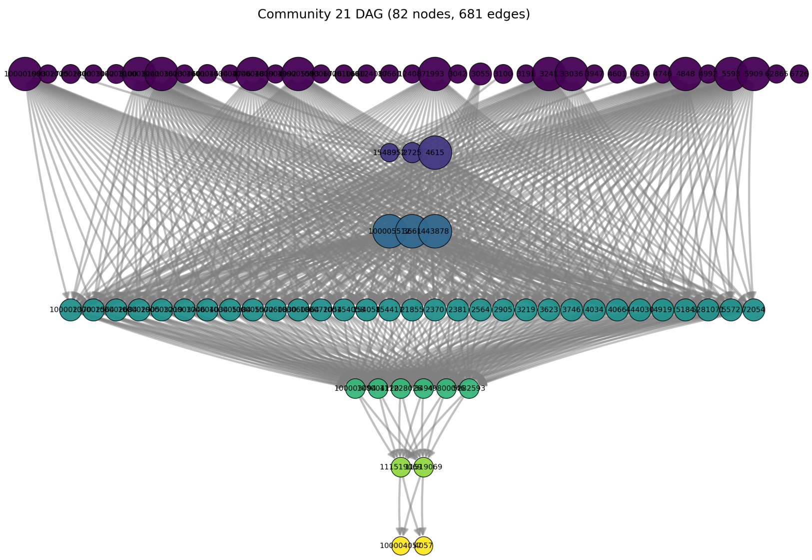

Figure 7: DAG visualization of community 21

6 Results

As regards the similarity network, the density value suggests a sparse but connected network, since there are communities with high density. The fact that only a small fraction of possible pairs (with both thresholds) exceeds the similarity threshold indicates that most drugs are not similar to the majority of the others, which is consistent with biological networks characterized by high specificity and few strong similarities.

The structure of the dataset exhibits a non-random modular organization, in which the distribution of community sizes is strongly skewed, with many communities composed of few drugs and one of very large size (community clique). An important detail to highlight is that community size is not determinant for density. Indeed, large communities may be cliques (community 5), moderately connected (community 21), or sparse (community 188). The global community parameters are strongly influenced by extremely compact modules in which there is high profile redundancy and therefore strong target overlap. The clustering coefficient is high, indicating that communities are not linear or chain-like and suggesting the existence of more compact internal blocks. A final aspect to emphasize is the high separation between communities, since only two of them are connected.

With regard to the gene–gene analysis (co-occurrence network), it is important to note once again that, since $component\_count = 1$ for every community, each co-occurrence network is completely connected; that is, all genes are reachable through at least one path. From the high density values, a strong redundancy of co-targeting also emerges in this case. Furthermore, at least 50% of the edges are supported by a number of drugs ≥ 2.

Communities with similar density values show very different Fiedler values, indicating that, as highlighted previously, density alone is not sufficient to describe the global cohesion of a community. Communities with low Fiedler values (135, 188, 192) are more easily separable and therefore exhibit the presence of weakly connected internal subgroups. Communities with intermediate–high Fiedler values are dense but display internal modularity (not rigidly homogeneous), whereas those with high Fiedler values are globally cohesive and represent real cohesion. The clique community, thanks to its high Fiedler value, is confirmed to be a clique community not only mathematically (from the calculation of density) but also spectrally cohesive.

The construction and analysis of the DAG with respect to Community 21 highlights the presence of an internal pharmacological hierarchy (numerous sources and a limited number of sinks). In this context, it can be observed that a significant fraction of the drugs exhibits maximal profiles, indicating the existence not of a single dominant profile, but of multiple independent roots. From the analysis of the degree ratio, it emerges that the distribution is strongly skewed, with the majority of nodes being peripheral and fewer nodes dominating the hierarchy. The final hierarchy discovered after constructing the DAG shows the presence of multiple roots, a dominant intermediate layer, and a few extremely specific terminals (branched hierarchy).

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